Results: hybrid PQC on the StarFive VisionFive 2

Measured results from the FOCAL "Post-Quantum Retrofit" hackathon work: legacy vs hybrid post-quantum TLS 1.3 on a StarFive VisionFive 2 (JH7110, 4x SiFive U74 @ up to 1.5 GHz), OpenSSL 3.5.9.

- **Legacy:** X25519 key exchange (+ RSA-3072 server authentication)
- **Hybrid:** X25519MLKEM768 key exchange (+ ECDSA P-256 and ML-DSA-44 dual-signature authentication on our own server)

Two scenarios are measured throughout: 1. **FOCAL's hackathon server** (IPv4), TLS handshake time. It supports the hybrid key exchange but authenticates with RSA-3072 only, so hybrid authentication can't be used there 2. **Own server** on the local network, end-to-end time including full hybrid authentication

All numbers on one page (best / median / worst for every approach): **`FINAL_COMPARISON_REPORT.md`** (PDF). Full story: **PROJECT_SUMMARY.md** (PDF); sections 10–14 cover the results in this folder. Source code: `../src/`; analysis scripts: `../tools/`.

Folder layout: one folder per approach

Each approach is self-contained: a detailed report (`.md` + `.pdf`), all raw measurements, statistics, logs and the scripts used, so each can be written up on its own.

```
results/
├── README.md / .pdf                      this index
├── FINAL_COMPARISON_REPORT.md / .pdf     all numbers on one page (best / median / worst)
├── 01_baseline_default_clock/            BASELINE: default CPU setting, legacy vs hybrid
├── 02_dynamic_cpu_freq/                  APPROACH 1: raise the minimum CPU clock
├── 03_bitmanip_zba_zbb/                  APPROACH 2: RISC-V Zba/Zbb bit-manipulation extensions
├── 04_multicore/                         APPROACH 3: classical and PQ halves on separate CPU
cores
└── 05_combined_all_approaches/           ALL APPROACHES COMBINED vs the baseline (final result)
```

Final result: all three approaches combined

Scenario	Baseline: legacy	Hybrid, baseline setup	Hybrid, all approaches combined	vs legacy baseline
Own server, full hybrid auth (ML-KEM + ECDSA + ML-DSA-44), end-to-end median	21.04 ms	28.06 ms	**14.47 ms**	**–31 % (20/20 blocks)**
Hackathon server, hybrid key exchange (X25519MLKEM768), handshake median	38.36 ms	43.76 ms	**35.62 ms**	**–7 % (17/20 blocks)**
Idle temperature	45.9 °C	45.9 °C	47.2 °C	+1.3 °C

COMBINED = minimum CPU clock 750 MHz + Zba/Zbb OpenSSL build + ML-DSA verified on a second core inside the TLS handshake; alternated block by block with the baseline setup, 1000 runs each, 0 failures. Against FOCAL's server only the first two apply (it uses RSA authentication, so there's no ML-DSA signature to verify). **The "no increase in end-to-end latency" target is met with margin, with software and settings only.** See `05_combined_all_approaches/combined_REPORT.md`.

Results per approach

Minimum CPU clock (dynamic CPU frequency)

Scenario	Baseline: legacy, default CPU	Hybrid, default CPU	Hybrid, min clock 750 MHz
Hackathon server, handshake (median)	42.17 ms	48.30 ms	**40.26 ms**
Own server, full hybrid auth, end-to-end (median)	21.30 ms	27.84 ms	**16.19 ms**
Idle temperature	44.7 °C	44.7 °C	**45.2 °C**

Raising only the minimum CPU clock (`ondemand` governor, `scaling_min_freq` = 750 MHz) lets hybrid PQC beat the legacy baseline for +0.5 °C. Under identical CPU settings hybrid remains slightly slower than legacy (+0.5 ms on the LAN, +3–4.5 ms over the internet).

RISC-V bit-manipulation extensions (Zba/Zbb)

Rebuilding OpenSSL with `-march=rv64gc_zba_zbb` (no code changes) speeds up SHA-3/SHAKE by 50 %, ML-KEM by 13–19 % and ML-DSA verification by 15 %, while classical algorithms are unchanged. End-to-end, full hybrid on the LAN drops from 28.15 to 26.57 ms (–5.6 %, faster in 20/20 blocks) at the default clock: a real gain, but about 7x smaller than the clock-policy change.

Multicore: classical and PQ halves on separate cores (feasibility)

Running the classical and post-quantum halves of each hybrid step on two cores at once speeds up the client-side crypto 1.5–1.65x at every clock setting (parallel faster in 90–100 % of iterations). At the default clock, the crypto of one hybrid handshake drops from 16.0 to 10.1 ms; combined with bitmanip and min 750 MHz it's 4.7 ms. **Part 2, end-to-end:** with the ML-DSA signature moved into the TLS 1.3 Certificate message and verified on a second core while OpenSSL verifies ECDSA, the full hybrid connection drops from 28.19 to **24.57 ms (–12.8 %, faster in 20/20 blocks)** at the default clock, more than halving the gap to legacy (+6.5 → +2.9 ms). The key exchange halves (inside OpenSSL) are still sequential.

Files

`../docs/PROJECT_SUMMARY.md`

Full project summary: setup, benchmarks, bottleneck analysis, migration guide, limitations. Section 10 = baseline at the default CPU clock; section 11 = dynamic CPU frequency, including the recommended setting. Sections 1–9 refer to earlier raw data that isn't included in this folder.

`01_baseline_default_clock/`: BASELINE (default CPU setting: `ondemand`, min 312.5 MHz)

File	Contents
`baseline_REPORT.md` / `.pdf`	Report: legacy vs hybrid, 1000 runs per mode, both scenarios, best / median / percentiles / worst
`baseline_raw_1000.csv`	All 4000 measurements: `scenario,mode,block,run,ms`
`baseline_stats_1000.txt`	Computed statistics
`baseline_bench_1000.log`	Per-block log: governor, temperature, negotiated group / ML-DSA verification checks
`scripts/bench_1000.sh`	Benchmark script used (runs on the board)

`02_dynamic_cpu_freq/`: DYNAMIC CPU FREQ (raise only the minimum CPU clock)

File	Contents
`dynamic_cpu_freq_REPORT.md` / `.pdf`	**Final report**: default vs `ondemand` + min 750 MHz, alternated block by block, 1000 runs per mode; includes the **RECOMMENDATION** section comparing all settings
`dynamic_cpu_freq_screen_REPORT.md` / `.pdf`	Screening of four settings (ondemand min 750 / 1250 MHz, schedutil min 312.5 / 750 MHz), 500 runs per mode
`dynamic_cpu_freq_confirm_raw.csv`	Confirmation run, all 8000 measurements: `config,scenario,mode,block,run,ms` (D0 = default, D1 = min 750 MHz)
`dynamic_cpu_freq_confirm_temps.csv`	Confirmation run temperatures: `config,phase,elapsed_s,temp_c`
`dynamic_cpu_freq_confirm_stats.txt`	Confirmation run statistics, incl. paired per-block comparison
`dynamic_cpu_freq_confirm.log`	Confirmation run per-block log and checks
`dynamic_cpu_freq_screen_raw.csv`	Screening run, all 8000 measurements (D1–D4)
`dynamic_cpu_freq_screen_temps.csv`	Screening run temperatures (idle + under load, every 5 s)
`dynamic_cpu_freq_screen_stats.txt`	Screening run statistics
`dynamic_cpu_freq_screen.log`	Screening run per-config log and checks
`scripts/bench_dyn.sh`, `scripts/bench_cfg_ab.sh`	Screening and alternating confirmation scripts (need sudo; restore the default CPU setting at the end)

`03_bitmanip_zba_zbb/`: BITMANIP (RISC-V Zba/Zbb bit-manipulation extensions)

File	Contents
`bitmanip_REPORT.md` / `.pdf`	**Report**: plain `rv64gc` vs `-march=rv64gc_zba_zbb` OpenSSL build; micro-benchmarks and 1000-run end-to-end comparison vs the baseline
`bitmanip_micro_fixed1500.txt`	Crypto micro-benchmarks (RSA, ECDSA, ML-DSA, ML-KEM, X25519, SHA-3, SHA-256) for std / opt / bitmanip builds, 2 passes at a fixed 1.5 GHz
`bitmanip_micro_ondemand_UNRELIABLE.txt`	First attempt under the `ondemand` governor; clock noise made it inconsistent. Kept for transparency, not used
`bitmanip_e2e_raw.csv`	End-to-end, all 8000 measurements: `config,scenario,mode,block,run,ms` (config = plain / bitmanip)
`bitmanip_e2e_stats.txt`	Statistics incl. paired per-block comparison and the baseline for reference
`bitmanip_e2e.log`	Per-block log, library linkage of each binary, negotiated group / ML-DSA verification checks
`scripts/bench_bitmanip.sh`, `scripts/micro_bitmanip.sh`, `scripts/micro_fixed.sh`	End-to-end and micro-benchmark scripts

`04_multicore/`: MULTICORE (classical and PQ halves on separate CPU cores)

File	Contents
`multicore_feasibility_REPORT.md` / `.pdf`	**Report**: one core vs two cores (new thread / persistent worker), 3 pairs, 3 clock settings x 2 builds, findings and limits
`multicore_feasibility_raw.csv`	All 10,800 measurements: `label,pair,mode,iter,us,cpu_classical,cpu_pq` (label = build_clock)
`multicore_feasibility_stats.txt`	Statistics: best / median / p95 / worst, time saved, speed-up, share of iterations where parallel won, core usage
`multicore_feasibility.log`	Run log (clock setting, build, temperature, failures)
`scripts/parverify_run.sh`	Run script (needs sudo; restores the default CPU setting)
`multicore_e2e_REPORT.md` / `.pdf`	**Part 2 report**: parallel hybrid authentication in the real TLS handshake (ML-DSA signature in a Certificate extension, verified on a 2nd core), legacy vs v1 vs v2 one core vs v2 two cores, 1000 runs each; security tests
`multicore_e2e_raw.csv`	All 4000 end-to-end measurements: `mode,block,run,tcp_ms,handshake_ms,e2e_ms,pq_verify_ms,wait_ms`
`multicore_e2e_stats.txt`	Statistics: best / percentiles / median / worst, time breakdown, paired per-block comparison, vs baseline
`multicore_e2e.log`	Run log: per-block verification checks, failures (0), temperature
`scripts/bench_multicore_e2e.sh`	End-to-end benchmark script (no sudo needed)

`05_combined_all_approaches/`: ALL THREE APPROACHES COMBINED (final result)

File	Contents
`combined_REPORT.md` / `.pdf`	**Report**: BASE (baseline setup) vs COMBINED (min 750 MHz + bitmanip + ML-DSA on a 2nd core), alternated block by block, 1000 runs per mode, both scenarios; target check, temperatures, how the approaches add up
`combined_raw.csv`	All 8000 measurements: `config,scenario,mode,block,run,ms`
`combined_temps.csv`	Idle and per-block temperatures: `config,phase,block,temp_c`
`combined_stats.txt`	Statistics: best / median / mean / p95 / worst, paired per-block comparisons, target check
`combined.log`	Run log: per-block verification checks, failures (0), temperatures
`scripts/bench_combined.sh`	Benchmark script (needs sudo; restores the default CPU setting)

Source code (`../src/`) and analysis scripts (`../tools/`)

File	What
`src/hybrid_server.c`	TLS 1.3 test server: RSA-3072 for legacy clients; ECDSA P-256 + ML-DSA-44/65 dual signature for hybrid clients, either bound to the TLS exporter (v1) or in a TLS 1.3 Certificate extension over the hello messages (v2)
`src/hybrid_client.c`	Benchmark client: pinned certificate + pinned ML-DSA key, end-to-end timing, `RAW=1` per-connection output, `TRACE=1` handshake timeline; options `certext` (v2) and `par` (verify ML-DSA on a second core)
`src/pqc_client.c`	Key-exchange client for the hackathon server (IPv4/IPv6, selectable group)
`src/sigbench.c`	Signature micro-benchmark (RSA, ECDSA, ML-DSA)
`src/parverify.c`	Multicore feasibility test: classical and PQ halves one after another vs on two cores
`tools/analyze_1000.py`, `analyze_dyn.py`, `analyze_pair.py`, `analyze_multicore.py`, `analyze_mc_e2e.py`	Statistics used for the reports
`tools/md2pdf.py`	Markdown to PDF conversion used for the PDFs

Method notes

- Every individual measurement is recorded; modes (and, in the confirmation run, CPU settings) alternate in blocks of 50 so both see the same network conditions.
- Every hybrid block was checked: X25519MLKEM768 negotiated and, on the own server, the ML-DSA-44 signature verified.
- 0 failed connections across all runs in this folder.
- Bitmanip builds only run on RISC-V CPUs that implement Zba and Zbb (the U74 does).
- Not measured: energy (temperature used as the proxy); minimum clocks of 500 and 625 MHz.

Source: README.md